

Geometric and organic forms and modelling

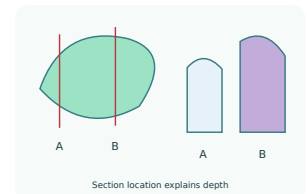
Organic geometry needs defined references as much as rectilinear furniture. Coordinate profiles and sections, distinguish surface appearance from construction, and inspect part interfaces before releasing fabrication data.

What you will accomplish

- Define profiles and section references
- Distinguish skin from structure
- Check parts and assembly orientation

Define profiles and section references

Establish floor datum, axes and view direction. Locate named sections on the plan where depth or form changes. A curved elevation does not reveal whether the body is a constant extrusion or a changing surface. Coordinate overall references across views rather than drawing each view independently.



Named section locations explain changing depth.

Distinguish skin from structure

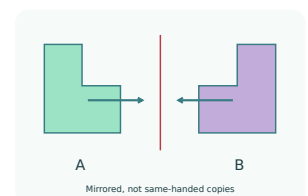
Compare a solid form, ribs with a skin, and an assembly of shaped sections. Show support paths, interfaces and finish build-up for each. These are design strategies requiring material and fabrication review, not bending or machine-setting instructions. Identical outer silhouettes can conceal different weights, edges and service access.



Matching silhouettes do not imply matching construction.

Check parts and assembly orientation

Number parts and define face, orientation, mating surface and insertion sequence. In the example A and B are mirrored ribs, not identical same-handed copies. Exporting both as A can change the interface. Check virtual assembly and interference, then state which uncertainties need a physical sample before specialist fabrication release.



Mirrored parts retain distinct identities.

Worked case and decision

The outer form is correct, but B was exported with A's handedness. Virtual assembly exposes a collision. Correct identity and orientation, then recheck the interface rather than enlarging the whole object to conceal the part error.

Decision: correct the mirrored part and recheck assembly.

Check understanding

Does a curved elevation define the complete depth geometry?

- No; coordinated sections are needed
- Always
- Shading is enough

Correct: No; coordinated sections are needed. A silhouette can conceal changing depth.

Lesson questions

Does virtual assembly replace every physical check?

No. Material and connection performance still need suitable verification.

Practice

Submit coordinated profiles, construction alternatives and an A/B assembly check.

References and sections

Skin and structure

Part identity and assembly

Review your work against the criteria. Saving it does not constitute technical project approval.

- Sections located on plan
- Structure distinguished
- Handedness and interference checked